

cultural replacement, drawing on ethnographic analogies with more recent hunter-gatherer groups who selectively adopted cultivation without abandoning foraging (Ikeya 2021). Set beside the isotopic and archaeobotanical evidence discussed above, this comparison across residue chemistry, ethnoarchaeology, and radiocarbon-dated crop remains points to a transition measured in generations rather than years: uneven across regions, and shaped as much by local ecology and existing social structures as by the simple availability of a new crop that migrating farmers brought with them.

So if rice did not immediately replace shellfish and other wild foods, what actually changed? The clearest shift is not in the raw list of foods people ate but in the temporal and social structure of subsistence itself. Shellfish collection, as the seasonal layering and caloric modeling from sites like Awazu and Isarago demonstrate, was organized around short-term, immediate returns: a bed of clams could be harvested within a day of locating it, and the resource replenished on a seasonal cycle largely outside human control (Habu et al. 2011). Rice cultivation demanded something different. Jomon and early Yayoi communities had to invest labor months before any return was realized, through field preparation, planting, weeding, and water management, traces of which survive archaeologically at Yayoi sites as paddy bunds, irrigation canals, and other water-control features. That shift toward delayed-return, storable production carries consequences well beyond diet. A stored rice surplus creates the possibility of unequal accumulation, and control over land and water becomes a new axis of social differentiation in a way that a naturally replenishing, widely accessible shellfish bed simply could not. Jomon society was not perfectly egalitarian to begin with: Kiriya and Kusaka (2017) documented variation in marine dietary dependence by burial cluster, sex, and mortuary ornamentation even within the Late and Final Jomon period at Yoshigo. But the later addition of